



PlateLoop

Know what's eaten. **Waste less.**

An AI food scanner that recognises every dish and measures what is served, eaten and left, down to the nutrients, so canteens cook the right amount and everyone who eats gets their own nutrition feedback.

ECOLOOP HACKATHON · SINGAPORE

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Class 3A15

1 The problem

755,000t

of food was thrown away in Singapore in 2023. **Only about 18% was recycled.**

28%

of the world's food waste comes from **food service**: canteens, restaurants and caterers.

8–10%

of global greenhouse gas emissions come from **food that is lost or wasted.**



Canteens **cook by guesswork**, because nobody measures what people actually eat.

WL

"I get the same big scoop as everyone, and I'm told to finish it."

Wei Ling, 15, student

ML

"I cook by feel. Some days half the kailan comes back."

Mrs Lim, canteen operator

NA

"By the time we notice a patient isn't eating, it's been days."

Nurse Aisha, Ward 7B

MT

"I want to build muscle, but I have no idea what the canteen gives me."

Marcus, office worker

2 Our process

design thinking, from lunch queue to prototype

1

Empathise

Watched how lunch really runs, mapped four people it affects and read the research on each.

2

Define

One problem: kitchens can't see what is eaten, dish by dish and person by person.

3

Ideate

Compared five ways to measure: bin weighing, surveys, photo apps, pre-orders and a tray scan.

4

Prototype

Modelled the scanner in 3D, built its kiosk, a kitchen dashboard, three apps and a 3D game.

5

Test

Walked every step until it broke, then turned each problem into a feature.

Face not recognised → type your class number

Clean-plate pressure → compared with your own usual

Alerts ignored → nurse actions in one tap

"Is it filming me?" → consent, no photos leave

Final prototype: PlateLoop Scanner

Scan your tray before you eat and again after. The AI recognises **every dish**, measures **how much of each** you ate, and works out the **nutrients** in it, in about two seconds.

620g

SERVED

–

95g

LEFT

=

525g

EATEN



Food type

the AI recognises each dish



Amount

grams served, left and eaten



Nutrients

Energy Protein Carbs Fat Fibre Sodium

Food groups

Proven method: research systems that scan trays before and after the meal identify ~89% of foods, and their nutrient estimates closely match the real values (Pfisterer et al., JMIR Aging 2022; Lu et al., 2020).

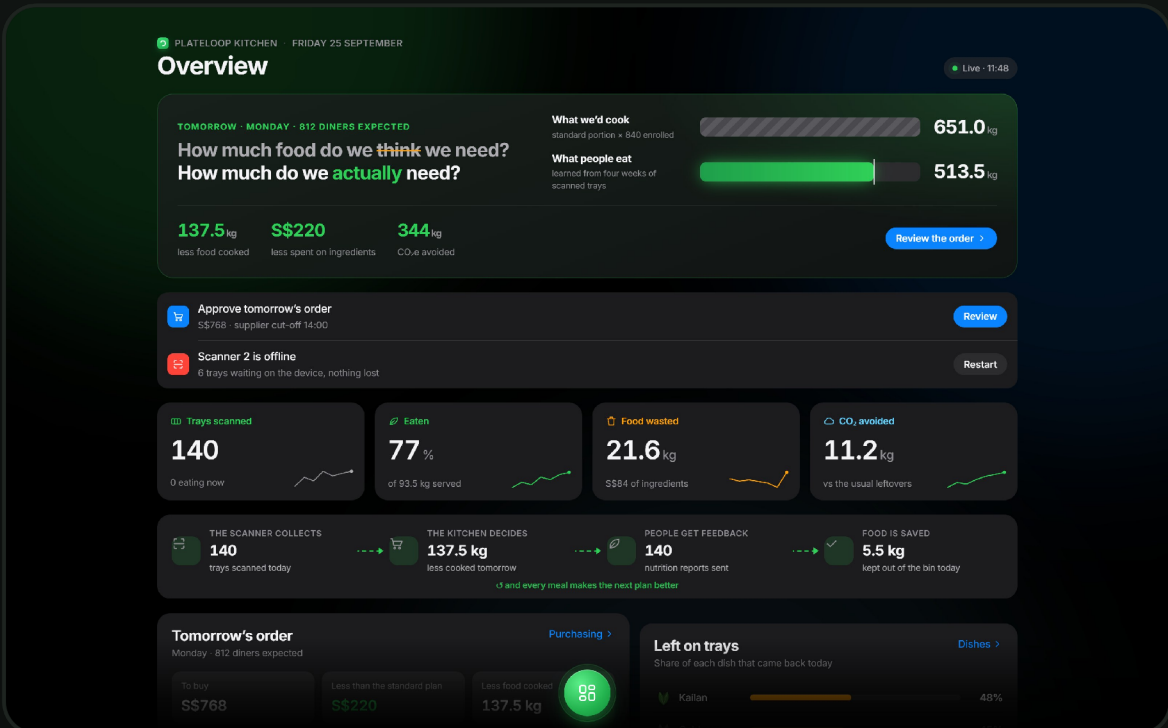
- Depth camera**
Finds each dish and measures its volume.
- Face camera**
Signs you in with a code, never a stored photo.
- 10.1" screen**
Steps, your result and what you ate.
- Weighing platform**
Checks the camera's estimate in real grams.
- Food waste bin**
Weighs the scraps, so nothing goes uncounted.



3D model, rendered in Blender

3 One scan, four apps

each built from research on the people who use it



PlateLoop Kitchen CANTEN STAFF

- ✓ **Tomorrow's order** from real attendance, weather and events.
- ✓ **Waste by dish**, so the kailan nobody eats gets smaller.
- ✓ **Carbon ledger**: CO₂ avoided, ready for reporting.

\$S28,500

ingredients saved a year in an 800-meal school (illustrative)

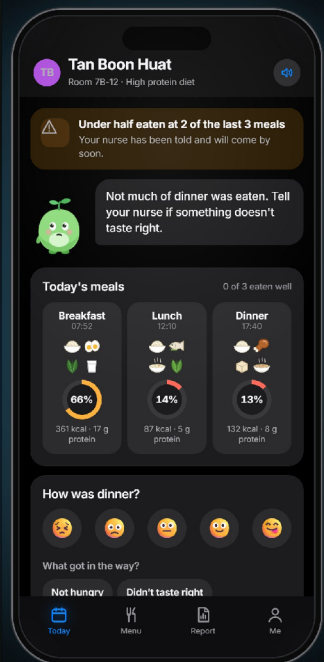


Loopi STUDENTS

- ✓ A **virtual pet** fed by the lunch you really ate.
- ✓ **Food groups**, not calories.
- ✓ **Class races**, streaks and challenges.

–35%

plate waste when classes weighed and posted their waste (Swedish school study, 2004)

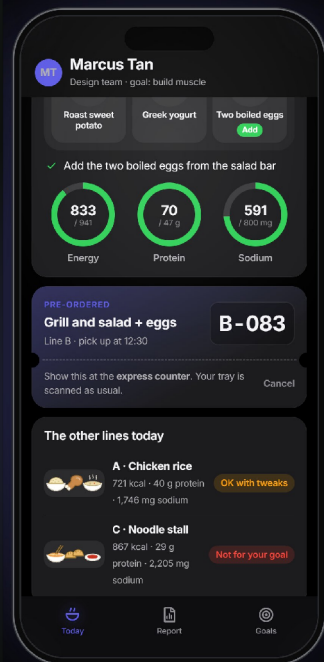


Loopi Care HOSPITALS

- ✓ **Every tray measured**, no food charts.
- ✓ Patients **choose tomorrow's meals**.
- ✓ Alerts reach the nurse **with the reason**.

29%

of inpatients were malnourished (Lim et al., Clinical Nutrition 2012)



Loopi Work OFFICES

- ✓ A **daily pick** for your health goal.
- ✓ **Pre-order** and skip the queue.
- ✓ **Private**: teams, never names.

6 in 10

Singaporeans eat out at least 4 times a week (HPB)

4 Impact

one 800-meal school, a year

5.7t

less food thrown away

14.3t

CO₂e avoided

5.1x

return on the subscription

2.4 mo

to pay back a year



Try it live

Bjiwoo.github.io/
plateloop

Evidence: 86 catering sites that measured their waste cut it 36% in a year, with a 6:1 return (Champions 12.3, 2018). Our estimate assumes a smaller 30% cut, 25% wasted today, \$S2.50 a meal. To validate in a pilot.